Six-stream cloud propagation: 500-event sample

NAUSICAA analysis

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Method

For each of two MC source samples, 500 events were propagated into three HDF5 clouds: positive ions, pure-Xe electrons and Xe–15% He electrons. One event location was sampled uniformly over the 0–150 cm drift length. Ions and electrons used complementary distances. The Julia launcher used six independent processes and separate output files. Timing includes Julia startup, source reads, per-carrier diffusion, histogramming and HDF5 writes.

Run results

Workers: 6. Events per stream: 500. Total wall time: 15.33 s. Total cloud storage: 310.3 MB.

Sample	Cloud	Time (s)	HDF5 (MB)	Peak RSS (GB)
bb0nu	ion	13.14	39.1	0.84
bb0nu	pure Xe electron	15.15	71.4	0.86
bb0nu	Xe–15% He electron	14.37	56.3	0.83
electrons	ion	12.65	30.0	0.85
electrons	pure Xe electron	14.72	64.8	0.86
electrons	Xe–15% He electron	13.71	48.7	0.82

The sum of the six individual peak resident-memory measurements is 5.05 GB; it bounds the simultaneous peak from above.

Output and checks

Each cloud file contains occupied Float32 histogram-bin positions in mm, Int64 carrier counts, explicit bin edges, common event location, both drift distances, evaluated diffusion widths, source identity and random seeds. The validation checks each event's three matching locations and produced carrier totals, the complementary drift distances, and the charge ledger between sampled, retained and out-of-range carriers.